17. Integration

J.P. Bastos*

EPH 1100: Analytical Tools

Reading: Hoy et al., §16.1–16.3.

- **§16.1:** Antiderivatives (indefinite integrals) and the basic integration rules.
- **§16.2:** The definite integral as the limit of a sum of rectangles; the fundamental theorem of calculus.
- **§16.3:** Properties of integrals; producer and consumer surplus.

1. Introduction

Throughout this course, we focused on going from totals to marginals. We started with a total (cost, revenue, utility) and took its derivative to study how *marginal changes* in these functions inform economic behavior.

Alternatively, often the marginal is what we know. For instance, a survey can tell us how much buyers are willing to pay for each additional unit of a good (the demand curve), but we may also want to know the total value across all buyers. If the marginal were constant, the total would be the marginal times the number of units. We would simply add how much each buyer is willing to pay.

However, when the marginal changes with every unit, we need to add up its value unit by unit. This running total is called an **integral**. Integration is the reverse of differentiation: differentiating a total gives the marginal, and integrating the marginal gives back the total.

2. Antiderivatives

Reversing differentiation means starting from a derivative and asking which function it came from.

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Definition. A function F is an **antiderivative** of f if its derivative is f:

$$F'(x) = f(x)$$

The **indefinite integral** of f is the set of all its antiderivatives:

$$\int f(x)\,dx = F(x) + C$$

where C is any constant, called the **constant of integration**. The function being integrated, f, is the **integrand**, and dx says that x is the variable we integrate over.

We need the constant because the derivative of a constant is zero, so differentiating a function erases any constant it contains. For instance, the functions

$$x^2 \qquad x^2 + 7 \qquad x^2 - 100$$

all have derivative $2x$, and knowing only the derivative, we cannot tell which of them we started from. In economics, the constant usually has a meaning. Since fixed cost does not change with output, it disappears when we differentiate total cost to get marginal cost. As a result, integrating marginal cost recovers total cost only up to the fixed cost, which we have to find from other information.

2.1 Integration Rules

Each integration rule below reverses a differentiation rule, and we can always check an integral by differentiating the answer, which should give back the integrand.

Rule 1: Power function

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \qquad \text{(for any power except } -1\text{)}$$

To integrate a power of x, we raise the power by one and divide by the new power, since differentiating the result brings the new power down and lowers the power by one, which undoes both steps.

Examples:

$$\int x^3\,dx = \frac{x^4}{4} + C$$

$$\int x^{1/2}\,dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}\,x^{3/2} + C$$

$$\int x^{-2}\,dx = \frac{x^{-1}}{-1} + C = -x^{-1} + C$$

To check the first one, differentiate the answer:

$$\frac{d}{dx}\left(\frac{x^4}{4} + C\right) = \frac{4x^3}{4} = x^3$$

Rule 2: Constant function

$$\int k\,dx = kx + C$$

The derivative of kx is the constant k, so integrating k gives back kx.

Example:

$$\int 5\,dx = 5x + C$$

Rule 3: Logarithmic rule (power of -1)

$$\int \frac{1}{x}\,dx = \ln x + C \qquad \text{(for positive } x\text{)}$$

The power rule does not work for a power of -1, because raising the power by one gives a new power of zero, and we cannot divide by zero. Instead, since the derivative of $\ln x$ is $\frac{1}{x}$, integrating $\frac{1}{x}$ gives back $\ln x$. Because the log is defined only for positive numbers, this rule holds only for positive x.[1]

Rule 4: Exponential function

$$\int e^x\,dx = e^x + C$$

Because the exponential function is its own derivative, it is also its own antiderivative.

Rule 5: Constant multiple

$$\int k\,f(x)\,dx = k\int f(x)\,dx$$

[1] For negative x, the antiderivative of $\frac{1}{x}$ is $\ln|x|$. We will not need this case.

A constant multiplying the function can be moved outside the integral, because a constant multiple carries through differentiation unchanged.

Examples:

$$\int 6x^2\,dx = 6 \times \frac{x^3}{3} + C = 2x^3 + C$$

$$\int \frac{4}{x}\,dx = 4\ln x + C$$

$$\int 3e^x\,dx = 3e^x + C$$

Rule 6: Sum and difference

$$\int \big[f(x) \pm g(x)\big]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

A sum is integrated term by term, just as it is differentiated term by term. Each term has its own constant of integration, but since the sum of constants is itself a constant, we write a single constant at the end.

Examples:

$$\int \big(3x^2 + 2x + 1\big)\,dx = x^3 + x^2 + x + C$$

$$\int \left(x^{1/2} - \frac{2}{x}\right)dx = \frac{2}{3}\,x^{3/2} - 2\ln x + C$$

To check the first one, differentiate the answer:

$$\frac{d}{dx}\big(x^3 + x^2 + x + C\big) = 3x^2 + 2x + 1$$

Rule 7: Reverse chain rule

$$\int f(ax+b)\,dx = \frac{1}{a}\,F(ax+b) + C$$

Here F is the antiderivative of f, and the inner function is linear,

$$u = ax + b$$

where a and b are constants and a is not zero.

To see where the rule comes from, start with a function and differentiate it with the chain rule:

$$\frac{d}{dx}(2x+1)^3 = 3(2x+1)^2 \times 2 = 6(2x+1)^2$$

The derivative of the inner function, 2, appears as an extra factor, along with the power 3 that the power rule brings down. Integrating $6(2x+1)^2$ should take us back to $(2x+1)^3$. We raise the power by one and divide by the new power, as in the power rule, and also divide by 2 to remove the extra factor from the inner function:

$$\int 6(2x+1)^2\,dx = 6 \times \frac{1}{2} \times \frac{(2x+1)^3}{3} + C = (2x+1)^3 + C$$

In general, the derivative of the inner function is a, so we integrate $f(u)$ as if u were x and divide by a. The rule works only when the inner function is linear, because only then is its derivative a constant that we can divide by. For a nonlinear inner function, such as the x^2+1 in $(x^2+1)^3$, the chain rule brings out $2x$, which depends on x, so we cannot remove it by dividing as we did with the 2. This particular function can be multiplied out and integrated term by term, as in the next subsection, but functions that cannot be rewritten this way need other methods, which this course does not cover.

Examples:

$$\int (3x+2)^4\,dx = \frac{1}{3} \times \frac{(3x+2)^5}{5} + C = \frac{(3x+2)^5}{15} + C$$

$$\int \frac{1}{2x+1}\,dx = \frac{1}{2}\ln(2x+1) + C$$

$$\int e^{3x}\,dx = \frac{1}{3}\,e^{3x} + C$$

In the second example, the outer function is $\frac{1}{u}$, whose antiderivative is $\ln u$ by Rule 3, so the answer holds only where $2x+1$ is positive.

A common mistake is to forget to divide by a. For instance, the integral of $\frac{1}{2x}$ is not $\ln(2x)$, whose derivative is $\frac{1}{x}$, but

$$\int \frac{1}{2x}\,dx = \frac{1}{2}\ln(2x) + C$$

The constant multiple rule gives $\frac{1}{2}\ln x + C$ instead, which is the same answer, since the two differ only by the constant $\frac{1}{2}\ln 2$.

Products and Quotients

Unlike differentiation, integration has no product or quotient rule, because no formula gives the integral of a product from the integrals of its factors. When a product can be multiplied out, or a quotient has a single term in the denominator and can be split into separate terms, we do that first and then integrate term by term. Products that cannot be rewritten this way, such as xe^x, call for a method called integration by parts, which this course does not cover.

Examples:

$$\int x(x+2)\,dx = \int \left(x^2+2x\right)dx = \frac{x^3}{3} + x^2 + C$$

$$\int \frac{x^3+2}{x}\,dx = \int \left(x^2+\frac{2}{x}\right)dx = \frac{x^3}{3} + 2\ln x + C$$

3. The Definite Integral

Notice that the antiderivative we just learned gives a function rather than a specific number. Often, however, we want a number, such as the variable cost of producing 30 units or the extra cost of going from 20 to 30 units. The definite integral is the tool that gives us these numbers, by evaluating the integral over a range.

Take a firm with marginal cost

$$MC(q) = 2q + 10$$

and suppose we want its variable cost of producing 30 units. We can approximate it by splitting the 30 units into 5 blocks of 6 units and pricing every unit in a block at the marginal cost of the block's first unit. The five blocks then cost

$$6\times\big[MC(0)+MC(6)+MC(12)+MC(18)+MC(24)\big] = 6\times(10+22+34+46+58) = 6\times 170 = 1{,}020$$

Each term is the area of a rectangle whose width is the 6 units in the block and whose height is the marginal cost of the block's first unit (Figure 1). This approximation is too low, because marginal cost rises within each block while we price every unit at the cost of the cheapest unit in its block. Using smaller blocks reduces the error. With 30 blocks of 1 unit, for instance, the height of each block is still the marginal cost of its first unit, so the sum is

$$1\times\big[MC(0)+MC(1)+\cdots+MC(29)\big] = 1{,}170$$

The table shows this sum together with a finer split into 300 blocks:

Number of blocks	Block width	Sum of rectangles
5	6	1,020
30	1	1,170
300	0.1	1,197

As the blocks get narrower, the rectangles fill more of the region under the marginal cost curve, and their sum approaches the area of that region, 1,200, which we compute exactly in the example below. This limit is called the definite integral.

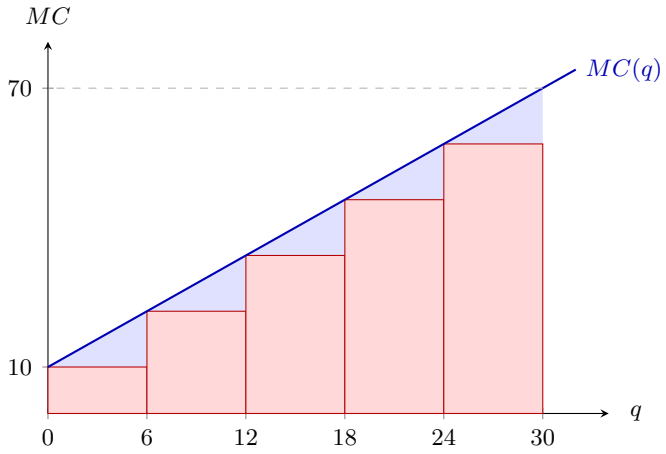


Figure 1: Approximating Variable Cost with Rectangles.
Notes: Each rectangle prices a block of 6 units at the marginal cost of the block's first unit; the five rectangles add up to 1,020. The blue region between the rectangles and the curve is the part they miss. The whole area under the curve from 0 to 30 units, rectangles and blue region together, is 1,200, the variable cost of producing 30 units.

Definition. The **definite integral** of f from a to b,

$$\int_a^b f(x)\,dx$$

is the limit of the sum of rectangles as their width shrinks to zero. When the function is positive, it is the area of the region under the curve and above the horizontal axis, between a and b. Where the function is negative, the area between the curve and the horizontal axis counts as negative.

The notation follows the rectangles: the integral sign is a stretched S, for sum, $f(x)$ is the height of each rectangle, dx is its width, and a and b, called the **limits of integration**, are where the sum starts and ends.

3.1 The Fundamental Theorem of Calculus

Fundamental Theorem of Calculus. If F is any antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

The difference is often written with brackets:

$$\big[F(x)\big]_a^b = F(b) - F(a)$$

In the cost example, marginal cost plays the role of f and total cost the role of F, so integrating marginal cost from 20 to 30 units gives the change in total cost between those two output levels. The same holds for any rate of change: integrating a country's net investment over time, for

instance, gives the change in its capital stock. Any antiderivative works here, because the constant of integration appears in both $F(b)$ and $F(a)$ and cancels when we subtract.

Example: For the firm with marginal cost

$$MC(q) = 2q + 10$$

find the variable cost of producing 30 units, and the extra cost of increasing output from 20 to 30 units.

Step 1: Find an antiderivative of marginal cost.

$$\int (2q + 10)\, dq = q^2 + 10q + C$$

Step 2: Find the variable cost of 30 units. By the fundamental theorem, the integral of marginal cost from zero to 30 is

$$\int_0^{30} MC(q)\, dq = TC(30) - TC(0)$$

Total cost at zero output is the fixed cost, so this difference is the variable cost of 30 units:

$$\int_0^{30} (2q + 10)\, dq = \left[q^2 + 10q\right]_0^{30} = (900 + 300) - (0 + 0) = 1{,}200$$

This is the number the rectangle sums in the table were approaching.

Step 3: Find the extra cost of going from 20 to 30 units, which is the integral from 20 to 30:

$$\int_{20}^{30} (2q + 10)\, dq = \left[q^2 + 10q\right]_{20}^{30} = (900 + 300) - (400 + 200) = 600$$

Although the last 10 units are only a third of the output, they account for half of the variable cost, because marginal cost rises with output.

4. Functions of Two Variables

4.1 Partial Integration

Just as we take a partial derivative, we can integrate a function of two variables with respect to one variable while treating the other as a constant. This is called **partial integration**, and it uses the same rules as integration in one variable. What changes is the constant of integration. Because the other variable is held fixed, the constant can be any function of that variable, since its partial derivative with respect to the variable we integrate over is zero. For a function of x and y,

$$\int f(x, y)\, dx = F(x, y) + C(y)$$

where F is a function whose partial derivative with respect to x is f.

Example: A consumer's marginal utilities are

$$\frac{\partial U}{\partial x_1} = \frac{x_2}{x_1} \qquad \frac{\partial U}{\partial x_2} = \ln x_1 + 1$$

Find her utility function.

Step 1: Integrate the marginal utility of good 1 with respect to x_1, treating x_2 as a constant. By the constant multiple rule and Rule 3,

$$U(x_1, x_2) = \int \frac{x_2}{x_1}\, dx_1 = x_2 \ln x_1 + C(x_2)$$

The constant may depend on x_2, because the marginal utility of good 1 tells us nothing about how utility changes with good 2.

Step 2: Find the function of x_2 using the marginal utility of good 2. Differentiate the result of Step 1 with respect to x_2 and set it equal to the marginal utility of good 2:

$$\frac{\partial U}{\partial x_2} = \ln x_1 + C'(x_2) = \ln x_1 + 1$$

The first term already matches the $\ln x_1$ in the marginal utility, so subtracting it from both sides leaves

$$C'(x_2) = 1$$

and, integrating with respect to x_2,

$$C(x_2) = x_2 + c$$

where c is an ordinary constant.

Step 3: Combine the two steps:

$$U(x_1, x_2) = x_2 \ln x_1 + x_2 + c$$

Adding a constant to a utility function does not change which bundles the consumer prefers, so we can set c to zero. Had we treated the constant in Step 1 as an ordinary number, we would have missed the x_2 term entirely.

4.2 Double Integration

In one variable, the definite integral adds up thin rectangles, each with a width dx and a height $f(x)$. In the cost example, the width was a block of units and the height was marginal cost, so the area of each rectangle was the cost of that block, and the area under the curve was the variable cost. A **double integral** does the same in two variables, with thin columns in place of rectangles, and gives the volume under a surface.

Suppose a firm wants to know how many consumers are in its target market: consumers aged 25 to 40 with incomes from \$50,000 to \$100,000. Consumers are spread over income I, in thousands of dollars, and age A, in years, and the function $n(I, A)$ gives how densely they are packed at each combination of income and age, in consumers per thousand dollars of income per year of age, or consumers per unit of income and age for short. Figure 2 plots this function as a surface. It is highest around middle incomes and middle ages, where consumers are most common, and falls toward the extremes.

Take a small range of incomes, of width dI, and a small range of ages, of width dA. This cell of the base has area dI times dA, and the height of the surface above it is $n(I, A)$, so the volume of the column, area times consumers per unit of income and age, is the number of consumers in that cell. Adding up the columns over the whole segment, and letting the cells shrink, gives the number of consumers in the segment, which is the volume under the surface:

$$\int_{25}^{40} \int_{50}^{100} n(I, A)\, dI\, dA$$

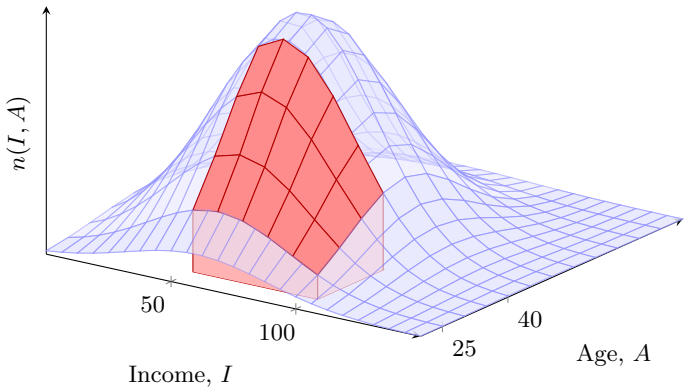


Figure 2: Consumers in a Market Segment.
Notes: The surface is the density of consumers by income, in thousands of dollars, and age, in years. The red rectangle on the base is the segment with incomes from \$50,000 to \$100,000 and ages from 25 to 40. The volume of the red solid above it is the number of consumers in the segment.

We will not compute double integrals in this course, but we should be able to read them. Each piece of the integral has a meaning:

$$\underbrace{\int_{25}^{40}}_{\substack{\text{ages} \\ \text{25 to 40}}} \quad \underbrace{\int_{50}^{100}}_{\substack{\text{incomes} \\ \text{\$50k to \$100k}}} \quad \underbrace{n(I, A)}_{\substack{\text{consumers per unit} \\ \text{of income and age}}} \quad \underbrace{dI\, dA}_{\substack{\text{small cell of} \\ \text{incomes and ages}}}$$

We read the integral from the inside out. The inner integral, over I, counts the consumers with incomes from \$50,000 to \$100,000 at a single age, holding A fixed as in partial integration. The outer integral, over A, then adds up these counts across ages 25 to 40. Integrating over A first and then over I gives the same number, so the order does not matter, as long as each pair of limits

stays with its own variable.

5. Applications

5.1 Total Cost from Marginal Cost

Integrating marginal cost gives total cost only up to the constant of integration, so to recover the total cost function we also need the fixed cost.

Example: A firm's marginal cost is

$$MC(q) = 3q^2 - 20q + 50$$

and its fixed cost is \$100. Find its total cost function.

Step 1: Integrate term by term, using the constant multiple and power rules on each term.

$$\int \left(3q^2 - 20q + 50\right) dq = 3 \cdot \frac{q^3}{3} - 20 \cdot \frac{q^2}{2} + 50q + C = q^3 - 10q^2 + 50q + C$$

Step 2: Find the constant. At zero output every term with q vanishes, so total cost equals the constant. Because total cost at zero output is just the fixed cost, the constant is 100:

$$TC(0) = C = 100$$

and total cost is

$$TC(q) = q^3 - 10q^2 + 50q + 100$$

To check, differentiate the total cost function, which should give back marginal cost:

$$\frac{d}{dq}\left(q^3 - 10q^2 + 50q + 100\right) = 3q^2 - 20q + 50$$

5.2 Consumer and Producer Surplus

The inverse demand curve gives, for each unit, the most that some buyer is willing to pay for it. Because the first units go to the buyers who value them most, willingness to pay falls as quantity rises. At the market price, however, every buyer pays the same amount, so the buyers of all but the last unit pay less than they would have been willing to. The gap between what a buyer is willing to pay and what she actually pays is her gain from the trade, and adding up these gaps over all the units sold gives **consumer surplus**.

Similarly, the inverse supply curve gives, for each unit, the least that some seller would accept to produce it, which for a competitive firm is the marginal cost of that unit. Adding up the gaps between the price and what sellers would accept gives **producer surplus**.

Consumer and Producer Surplus. With inverse demand $D(Q)$, inverse supply $S(Q)$, and equilibrium price P^* and quantity Q^*:

$$CS = \int_0^{Q^*} \big[D(Q) - P^*\big]\, dQ \qquad\qquad PS = \int_0^{Q^*} \big[P^* - S(Q)\big]\, dQ$$

Consumer surplus is the area below the demand curve and above the price. Producer surplus is the area above the supply curve and below the price.

When demand and supply are straight lines, the two areas are triangles and we do not need integrals to find them. When a curve bends, however, the area is no longer a triangle, and we need the integral to compute it.

Example: In the market for a medical device, inverse demand and inverse supply are

$$D(Q) = 200 - Q^2 \qquad\qquad S(Q) = 10Q$$

where quantity is in thousands of units and the price is in dollars. Find the equilibrium, consumer surplus, and producer surplus.

Step 1: Find the equilibrium. Set the two prices equal and collect the terms on one side:

$$\begin{aligned} 200 - Q^2 &= 10Q \\ Q^2 + 10Q - 200 &= 0 \\ (Q + 20)(Q - 10) &= 0 \end{aligned}$$

Quantity cannot be negative, so

$$Q^* = 10 \qquad\qquad P^* = S(10) = 10 \times 10 = 100$$

Step 2: Find consumer surplus. The gap between willingness to pay and the price is

$$D(Q) - P^* = 200 - Q^2 - 100 = 100 - Q^2$$

Integrate it from zero to the equilibrium quantity:

$$CS = \int_0^{10} \big(100 - Q^2\big)\, dQ = \left[100Q - \frac{Q^3}{3}\right]_0^{10} = 1{,}000 - \frac{1{,}000}{3} \approx 666.67$$

Quantities are in thousands, so consumer surplus is about \$667,000.

Step 3: Find producer surplus. The gap between the price and what sellers would accept is

$$P^* - S(Q) = 100 - 10Q$$

Integrate it from zero to the equilibrium quantity:

$$PS = \int_0^{10} \left(100 - 10Q\right) dQ = \left[100Q - 5Q^2\right]_0^{10} = 1{,}000 - 500 = 500$$

Since supply is a straight line, producer surplus is a triangle, and we can check the integral with the triangle formula:

$$\frac{1}{2} \times 10 \times 100 = 500$$

Quantities are in thousands, so producer surplus is \$500,000.

Had we treated demand as a straight line, consumer surplus would be the triangle with corners at (0, 200), (0, 100), and (10, 100):

$$\frac{1}{2} \times 10 \times (200 - 100) = 500$$

The integral gives a consumer surplus a third larger than this triangle, because the demand curve is concave (Figure 3): willingness to pay stays high over the first several thousand units and drops off only near the equilibrium quantity, so the region under the curve is larger than the triangle.

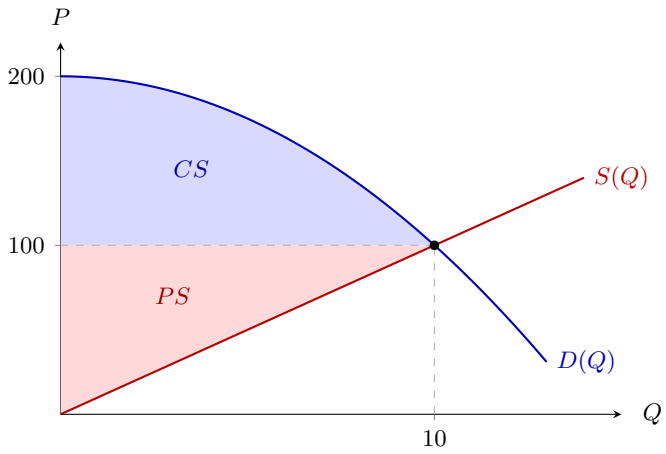


Figure 3: Consumer and Producer Surplus with a Curved Demand.
Notes: Inverse demand is $D(Q) = 200 - Q^2$ and inverse supply is $S(Q) = 10Q$, with equilibrium at 10 units and a price of 100. Consumer surplus (blue) is the area below demand and above the price, about 666.67. Producer surplus (red) is the triangle above supply and below the price, 500.

5.3 The Cost of a Price Increase

Recall that, by Shephard's lemma, the derivative of the expenditure function with respect to the price of a good is the Hicksian demand for that good:

$$\frac{\partial E}{\partial p_1} = x_1^H$$

Holding the price of good 2 and the target utility fixed, the expenditure function depends only on the price of good 1. By the fundamental theorem, integrating the Hicksian demand between two prices then gives the change in expenditure, which is the extra income a consumer needs to stay as well off when the price rises from p_1^A to p_1^B:

$$E(p_1^B, p_2, \bar{U}) - E(p_1^A, p_2, \bar{U}) = \int_{p_1^A}^{p_1^B} x_1^H(p_1, p_2, \bar{U})\, dp_1$$

Because we integrate over the price, which is on the vertical axis, each thin rectangle is a horizontal strip whose height is a small price change and whose length is the quantity she buys at that price. The integral is therefore the area to the left of the Hicksian demand curve between the two prices (Figure 4).

Example: A consumer's Hicksian demand for good 1 is

$$x_1^H = \bar{U}\left(\frac{p_2}{p_1}\right)^{0.5}$$

Suppose her target utility is 10, good 2 costs \$4, and the price of good 1 rises from \$1 to \$4. How much extra income does she need to stay as well off?

Step 1: Write the Hicksian demand as a function of the price of good 1 alone, by substituting the target utility of 10 and the price of 4 for good 2:

$$x_1^H = 10 \times \left(\frac{4}{p_1}\right)^{0.5} = 10 \times \frac{2}{p_1^{0.5}} = 20\, p_1^{-0.5}$$

Step 2: Find the antiderivative. By the power rule, and since dividing by 0.5 is the same as multiplying by 2,

$$\int 20\, p_1^{-0.5}\, dp_1 = 20 \times \frac{p_1^{0.5}}{0.5} + C = 40\, p_1^{0.5} + C$$

Step 3: Evaluate it between the two prices:

$$\int_1^4 20\, p_1^{-0.5}\, dp_1 = \left[40\, p_1^{0.5}\right]_1^4 = 40 \times 2 - 40 \times 1 = 40$$

She needs \$40 of extra income.

We can check this answer against two simple estimates. At the old and the new price she buys

$$x_1^H(1) = 20 \times 1^{-0.5} = 20 \qquad x_1^H(4) = 20 \times 4^{-0.5} = 10$$

Charging the \$3 price increase on all 20 units would give \$60, which overstates the cost, while charging it on only 10 units would give \$30, which understates it. Because her quantity falls from 20 to 10 as the price rises, the exact cost of \$40 lies between the two.

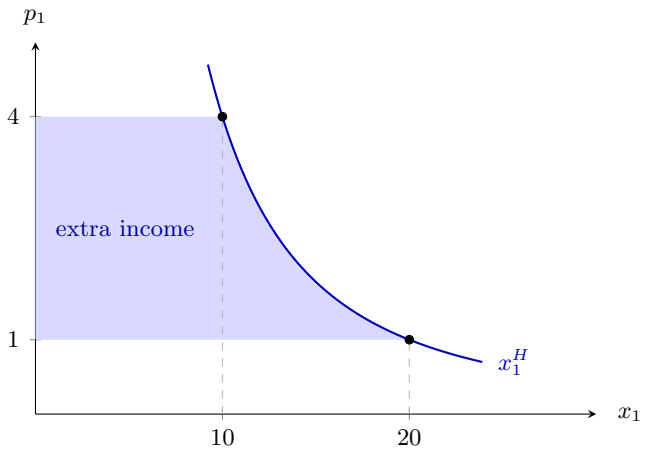


Figure 4: The Cost of a Price Increase.
Notes: Hicksian demand for good 1 at a target utility of 10 and a price of 4 for good 2, with the price of good 1 on the vertical axis. The shaded area, between the two prices and to the left of the curve, is the extra income the consumer needs when the price rises from 1 to 4, which is 40.

Practice Problems

1. Find the following indefinite integrals. Check each answer by differentiating.

   (a) $\int (4x^3 - 6x + 2)\,dx$

   (b) $\int \left(x^{1/2} + 3x^{-2}\right) dx$

   (c) $\int \left(5 + \frac{2}{x}\right) dx$

   (d) $\int \left(e^{4x} + \frac{3}{2x+1}\right) dx$

   (e) $\int (x+1)(x-3)\,dx$

   (f) $\int \frac{2x^2 - 3}{x}\,dx$

2. Evaluate the following definite integrals.

   (a) $\int_0^3 (2x+1)\,dx$

   (b) $\int_1^4 (3x^2 - 2x)\,dx$

   (c) $\int_1^9 x^{-1/2}\,dx$

   (d) $\int_1^e \frac{1}{x}\,dx$

   (e) $\int_0^1 e^{2x}\,dx$

3. A firm's marginal cost is $MC(q) = q^2 - 4q + 10$, and its fixed cost is \$100.

   (a) Find the total cost function.

   (b) Find the total cost of producing 6 units.

   (c) Find the extra cost of increasing output from 6 to 9 units, using a definite integral. Check it with your answer to (a).

   (d) At what output is marginal cost lowest?

4. A firm's marginal products of labor and capital are

$$\frac{\partial Q}{\partial L} = 2K \qquad \frac{\partial Q}{\partial K} = 2L + 3$$

   and it produces nothing when it uses no inputs. Find its production function.

5. The amount of oil underground varies across a region. Let x be the distance east of a reference point and y the distance north, both in miles, and let $r(x, y)$ be the oil underground at each location, in barrels per square mile. A company leases the block of land from 2 to 5 miles east and from 0 to 3 miles north of the reference point. Write the double integral that gives the amount of oil under the company's lease.

6. A city uses electricity at a rate that depends on the hour of the day h, from 0 to 24, and the day of the year t, from 0 to 365. The function $e(h, t)$ gives this rate, in megawatt-hours per hour of the day per day of the year. Explain in words what the following integral measures, and what each of its pieces means:

$$\int_0^{31} \int_{17}^{21} e(h, t)\, dh\, dt$$

7. A country's net investment (the change in its capital stock per year) is $I(t) = 30\, t^{0.5}$, in billions of dollars, where t is years from today. Today's capital stock is \$500 billion.

   (a) By how much does the capital stock grow over the next 4 years?

   (b) What is the capital stock 9 years from today?

8. A market has inverse demand $D(Q) = 80 - Q$ and inverse supply $S(Q) = 20 + Q$.

   (a) Find the equilibrium price and quantity.

   (b) Compute consumer surplus and producer surplus with integrals, and check each with the triangle formula.

   (c) A tax of \$10 per unit is imposed. Find the new quantity, the price buyers pay, and the price sellers keep. (Hint: the price buyers pay exceeds the price sellers keep by the tax.)

(d) The deadweight loss of the tax is the surplus lost on the units that are no longer traded. It equals the integral of $D(Q) - S(Q)$ between the new and the old quantity. Compute it.

9. Inverse demand for a good is $D(Q) = 64 - Q^2$, and the market price is \$28.

   (a) How many units are sold?

   (b) Compute consumer surplus.

   (c) Compare your answer with the triangle that has the same corners. Which is larger, and why?

10. A competitive firm sells output at $p = 20$. Its marginal cost is $MC(q) = 0.5q + 5$.

    (a) Find the profit-maximizing output.

    (b) Compute producer surplus, the area between the price and the marginal cost curve from zero to the optimal output.

    (c) Use an integral to find the variable cost of producing the optimal output. Show that producer surplus equals revenue minus variable cost.

11. A consumer's Hicksian demand for heating oil, at her current utility and the prices of other goods, is $x^H = 1{,}200\, p^{-0.5}$, in gallons per year.

    (a) The price rises from \$1 to \$9 a gallon. Estimate the extra income she needs to stay as well off by multiplying the quantity she buys at \$1 by the price increase.

    (b) Find the exact amount by integrating her Hicksian demand.

    (c) Explain why the two answers differ.